



BULUT EMRE SAKARYA

Software Engineer | AI & Automation · Backend · Mobile

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PROFESSIONAL SUMMARY

Software Engineer with 5+ years of experience building end-to-end products for the logistics and real estate industries, delivered under a **SaaS model**. I build autonomous **AI agents** (LangChain/LangGraph), high-scale **web scraping** pipelines, **React Native mobile apps**, and real-time **BI dashboards**. Two of my products are **live on Google Play and the App Store**, and every system I ship runs 24/7 on **Linux VDS** infrastructure I manage myself, with no dependency on no-code platforms. I design architectures that automate **90%** of manual workflows, enforce SLAs automatically, and apply **FinOps** principles to LLM cost control — one system generated **+1.5M TL in net revenue within 4 months**.

TECHNICAL SKILLS

- Languages:** Python, JavaScript, SQL, HTML/CSS
- Backend & API:** FastAPI, Node.js, RESTful API design, webhook management, async queue architecture, OTP authentication
- AI & Orchestration:** LangChain / LangGraph (stateful multi-agent workflows), OpenAI API, structured data extraction (Pydantic/JSON Schema), prompt engineering, LLM cost optimization
- AI-Assisted Development:** Claude (Claude Code), ChatGPT, Gemini, Replit Agent — context management across large codebases, rapid prototyping, code review and refactoring; disciplined verification and testing of AI output before it ships
- Mobile:** React Native / Expo (EAS Build), push notifications, Google Play & App Store release pipelines, subscription / RTDN integration
- Data & BI:** PostgreSQL (PostGIS), SQLite (WAL Mode), Supabase, Drizzle ORM, ETL & data cleaning, dashboard development with React + TanStack Query
- Web Scraping:** Selenium, requests/aiohttp, anti-bot evasion, session & rate-limit management
- Cloud & DevOps:** AWS (Certified Cloud Practitioner, CLF-C02), Docker, Linux server administration (Ubuntu VDS, Oracle Cloud), PM2, Cloudflare Tunnel, Git & GitHub
- Languages (spoken):** Turkish (native), English (B2 — AWS certification exam completed in English), German (A1, beginner)

PROFESSIONAL EXPERIENCE

FREELANCE / KARAARSLAN DIGITAL | Software Engineer (AI, Backend & Mobile)

11/2025 – Present

- Build **AI agents**, **ETL pipelines**, and **mobile applications** for SMEs and institutional clients, owning the full technical delivery from architecture design through store release.
- Tırdaş** (02/2026 – Present): Built an AI-powered logistics SaaS platform, WhatsApp dispatch bot, and mobile app, now **live on Google Play and the App Store**. Delivered the core build in 06/2026 and continue to run the platform under an **ongoing paid maintenance, feature, and support retainer**.
- Quadmate**: Shipped a location-based gaming-partner matchmaking app end to end (React Native + Supabase/PostGIS).
- Delivered corporate web, hosting, and SEO projects (CleanHome, Nilbaşak) and productized a resellable **"Ready Solutions"** line (AI customer agent, QR menu, KPI dashboard, data harvesting).
- Impact**: Automated **90%** of clients' manual data entry operations, compressing an **8-hour weekly workload down to 5 seconds**.

RE/MAX TRUE | AI & Automation Programmer

09/2024 – 10/2025

- Joined as a real estate agent; internal automations I wrote to streamline field operations were noticed by management, and I moved into the **AI & Automation Programmer** role within two months (11/2024).
- Solely architected the **"TrueMax"** ecosystem — an anti-ban scraper, an SLA lead dispatcher, and an AI outreach orchestrator — and redeployed the same engine as **RED-i** for RedPartners.
- Expanded the office's active client portfolio by **60%+** and total branch revenue by **40%+**, generating **+1.5M TL in net revenue within 4 months**.
- Designed and built the **real-time BI dashboard** tracking 40+ agents, cutting weekly reporting time from 8 hours to 5 seconds.

ÖZGELECEK İK TEKNOLOJİLERİ A.Ş. | Data & Software Developer

04/2024 – 08/2024

- Developed **Python** automations for data processing and reporting, enabling the team to consistently exceed weekly output targets.
- Promoted from intern to Data & Software Developer within two months** based on technical delivery.

FREELANCE | Python Developer

03/2020 – 01/2024

- Built custom backend **automation scripts** and **web scrapers** for e-commerce clients, and maintained the systems I delivered.

EDUCATION

ANADOLU UNIVERSITY | B.S. in Management Information Systems (Open Education)

2025 – Present

- Honor Certificate — Spring Semester, 2025–2026 Academic Year

DOKUZ EYLÜL UNIVERSITY | Undergraduate Studies in Physics (partial coursework)

2021 – 2023

CERTIFICATIONS

- AWS Certified Cloud Practitioner (CLF-C02) — Amazon Web Services (10/2025)
- Information System Design Certificate — Anadolu University (05/2026)
- JavaScript Course — Boğaziçi Enstitüsü (2026)
- IT Internship — TNC Group · Europäisches Bildungsinstitut (07/2025)
- BTK Akademi — AI Ethics, Customer Insight, Communication & Network Management (05/2025)

PROJECTS

TIRDAŞ — AI-POWERED LOGISTICS SAAS, MOBILE APP & WHATSAPP BOT — Software Engineer (Solo)

An autonomous dispatching ops-center connecting logistics carriers with truck drivers. The moment a driver writes “I’m in Gebze, heading to Ankara, find me a load,” the system matches the best available freight, reports route weather, logs expenses, and escalates unresolved cases to a live support operator. The **mobile app is live on Google Play and the App Store**, sharing a single listing pool with the WhatsApp bot. [Python](#) · [FastAPI](#) · [LangGraph](#) · [PostgreSQL](#) · [React Native \(Expo\)](#) · [Meta Cloud API](#) · [PM2](#)

- **Colloquial NLP Pipeline:** Built a LangGraph & JSON Schema workflow translating slang and misspelled trucking terminology (e.g., “Kırkayak” → 10-wheel truck) into strict, structured database queries.
- **Geographic Validation Layer:** Wrote a validation layer normalizing free-text listings against an **81-province / 973-district** reference set; accompanying panel optimizations delivered a **300% speed-up**.
- **FinOps Budget Engine:** Engineered an ApiManager module that calculates real-time token spend per user and throttles the system on reaching the monthly cost limit (monthly_cost_limit).
- **Mobile & Backend:** Delivered the React Native (Expo) client, FastAPI backend, OTP authentication, push notifications, admin panel, and Google Play subscription/RTDN integration end to end.
- **Async Message Queue:** Built a self-healing PM2 queue that ingests high-frequency Meta Webhook bursts without payload loss during server outages.

TRUEMAX.AI — AI LEAD GENERATION & OUTREACH PIPELINE — AI Automation Engineer (Solo)

A revenue engine that scrapes property listings, reaches owners with human-like WhatsApp outreach, and auto-distributes warm leads to agents. **30,000+ property owners reached, 2,778 collaboration approvals** secured, across a scanned portfolio worth **39.1 billion TL**. Generated **+1.5M TL in net revenue** for the office in 4 months. [Python](#) · [Selenium](#) · [Node.js](#) · [WhatsApp](#) · [LLM](#) · [SQLite](#)

- **Anti-Ban Hybrid Scraper:** Designed a hybrid extraction engine capturing a real browser session (cookie/session) via Selenium, then offloading heavy traffic to fast requests.Session tunnels.
- **Human-Simulation Algorithm:** Bypassed bot detection with dynamic sendSeen() hooks and randomized 2–5 second jitter delays.
- **Strict 120-Minute SLA Engine:** Wrote the accountability module that assigns warm leads to agents and reassigns them Round-Robin to the next agent if uncontacted within 2 hours.
- **Cost Optimization:** Cut LLM spend by ~70% through prompt compression and caching, and redeployed the engine as **RED-i** for RedPartners.

REAL-TIME BI & KIOSK DASHBOARD — Frontend & Data Architect

A 24/7 lobby TV broadcast and management panel tracking real-time sales competition across **40+ agents**, office revenue, and live currency/gold/crypto rates; ran in production on **215 real transactions**. [React](#) · [Express](#) · [SQLite \(WAL\)](#) · [TanStack Query](#) · [Google Sheets API](#)

- **Data Sanitization Layer:** Built RegEx pipelines normalizing manually entered malformed values (e.g., “1.500.00tl”, “1500000”), cutting the reporting error rate from **15% to 0.01%**.
- **Lock-Free Concurrency:** Enabled SQLite WAL (Write-Ahead Logging) mode to prevent SQLITE_BUSY deadlocks during concurrent writes from two background bots.
- **Autonomous Kiosk Engine:** Coded a zero-refresh UI triggering the lobby browser via the DOM Fullscreen API and keeping data live with TanStack Query.

QUADMATE — LOCATION-BASED GAMING PARTNER MATCHMAKING APP — Software Engineer (Solo)

A mobile app helping players find the missing participant for a table or event nearby — users browse open tables on a map, join listings, or open their own table. [React Native \(Expo\)](#) · [Supabase](#) · [PostGIS](#) · [Push Notifications](#)

- **Geospatial Matching:** Implemented radius-based PostGIS queries, row-level security (RLS), and real-time subscriptions on Supabase.
- **Mobile Delivery:** Shipped iOS and Android from a single codebase, owning the push notification pipeline and store release process.

ADDITIONAL DELIVERIES

- **Crypto Strategy Bot:** A scheduled strategy bot and control panel written using only the Python standard library — leverage, TP/SL, and position sizing controls with a live PnL feed; demoed to the client in paper-trading mode.
- **Telegram Ops Bot:** Deployed a notification bot on Oracle Cloud that monitors defined sources 24/7 and surfaces one-tap inline-keyboard actions when conditions are met.
- **Corporate Web & Hosting:** Delivered corporate sites for CleanHome and Nilbaşak with SSL, quote forms, SMTP/HTTP mail integration, and SEO foundations — running uninterrupted since delivery.

All interactive demos and technical case studies: [bulutemresakarya.github.io](#)